Fig. 14 shows some representative thermocouples. The responsivity of a single thermocouple is very low and therefore to increase the responsivity, several junctions are connected in series to form a thermopile (Fig. 13). Thermopiles are series combination of around 20-200 thermocouples. Spectral response of thermocouples and thermopiles extends into the far-infrared band up to 40µm. These are suitable for measurements over a large temperature range up to 1800K. However, thermocouples are less suitable for applications where smaller temperature differences need to be measured with great accuracy, such as 0-100°C measurements with 0.1°C accuracy. For such applications, thermistors and RTDs are more suitable.

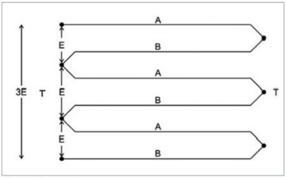

Fig. 13: Thermopile

Fig. 14: Thermocouples

Bolometers

Bolometers are the most popular type of thermal sensors. The sensing element in bolometers is a resistor with high temperature coefficient. While in photoconductors direct photon-electron interaction causes a change in the conductivity of the material, in bolometers the increased temperature and temperature coefficient of the element cause the resistance change.

Bolometers can be further categorised as metal bolometers, thermistor bolometers and low-temperature germanium bolometers.

Metal bolometers uses metals such as bismuth, nickel and platinum with temperature coefficient in the range of 0.3-0.5 per cent/°C. Thermistor bolometers are the most popular and find applications in burglar alarms, smoke sensors and other similar devices. Their sensor is a thermistor, which is an element made of manganese, cobalt and nickel-oxide. Thermistors have high temperature coefficient of up to 5 per cent/°C. The temperature coefficient varies with temperature as $1/T^2$. Thermistors are classified as negative-temperature-coefficient (NTC) and positive-temperature-coefficient depending upon whether their temperature coefficient of resistance is negative or positive.

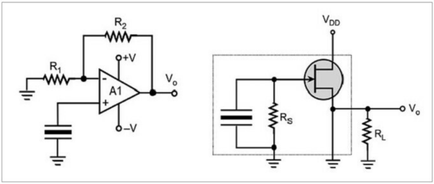

Fig. 15: Voltage mode operation of a pyroelectric sensor

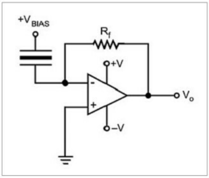

Fig. 16: Current mode operation of pyroelectric sensor

Pyroelectric sensors

Pyroelectric sensors are characterised by spontaneous electric polarisation, which is altered by temperature changes as light illuminates them. These are low-cost, high-sensitivity devices that are stable against temperature variations and electromagnetic interference. Pyroelectric sensors only respond to modulating light radiation and produce no output for a CW incident radiation.

Pyroelectric sensors operate in two modes: voltage and current. In voltage mode (Fig. 15), voltage generated across the entire pyroelectric crystal is detected. In current mode (Fig. 16), current flowing on and off the electrode on the exposed face of the crystal is detected. Voltage mode is more commonly used than current mode.

Voltage-mode pyroelectric sensors are generally integrated with a field-effect transistor. A shunt resistor (R_S) in the range of 10^{10} to 10^{11} ohms is added to provide thermal stabilisation. External connections include a power supply and load resistor R_L. The output voltage appears across R_L. Modulation frequency in current-mode operation can be much higher than in voltage-mode operation. Hence it is much easier to separate the signal from the ambient temperature drift.

To be continued...